



Jânio Anselmo, M.Sc.

Electrical Engineer (CREA-SC 074576-3) · PhD Candidate in Electrical Engineering – UFSC

Profile

Electrical engineer and M.Sc. in Electrical Engineering (concentration area: Biomedical Engineering) from UFSC, currently a PhD candidate at the same institution, researching mathematical modeling and experimental validation of pulsed electric fields (PEF). Co-founder and R&D Director of ENSA Tecnologia since 2009, leading a team of four in the development of microcontroller-based electronic systems, combining *hardware* design, signal processing and numerical modeling. Named inventor on two invention patent applications (2025 and 2026), with experience in scientific output, peer review of articles, thesis committees and teaching.

Areas of Interest

Electrical Engineering, Biomedical Engineering, Electronics, Microcontrollers, Digital Signal Processing, Image Processing, Cell Membrane Electroporation, Pulsed Electric Fields (PEF), Food Preservation, Microorganism Inactivation, Numerical Modeling, Business Management.

Academic Background

- 2023–2027 **PhD in Electrical Engineering (*ongoing*)**, *Federal University of Santa Catarina (UFSC)*, Florianópolis/SC
Graduate Program in Electrical Engineering (PPGEEL), *Stricto Sensu*, concentration area: Biomedical Engineering. Thesis: *Pulsed electric fields in continuous flow: effective electric dose modeling and experimental validation*. Advised by Prof. Daniela Ota Hisayasu Suzuki, Ph.D., and co-advised by Prof. Guilherme Brasil Pintarelli, Ph.D.
- 2015–2018 **Postgraduate Specialization in Business Management**, *Faculdade Municipal de Palhoça (FMP)*, Palhoça/SC
Lato Sensu specialization in business management.
- 2012–2014 **M.Sc. in Electrical Engineering**, *Federal University of Santa Catarina (UFSC)*, Florianópolis/SC
Stricto Sensu graduate program, concentration area: Biomedical Engineering.

Palhoça/SC – Brazil

☎ +55 (48) 99601-1213 • ✉ anselmo.janio@gmail.com

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🆔 0000-0002-6872-5089 • *Brazilian, born 03/05/1983, married*

- 2008–2015 **Technologist Degree in Telecommunications Systems**, *Federal Institute of Santa Catarina (IFSC)*, São José/SC
Undergraduate technologist degree in Telecommunications Systems.
- 2006–2011 **B.Eng. in Electrical Engineering – Telematics**, *University of Southern Santa Catarina (UNISUL)*, Palhoça/SC
Degree in Electrical Engineering with emphasis on Telematics and electives in power electronics and computer networks.
- 1997–2001 **Technical Diploma in Telecommunications**, *Federal Institute of Santa Catarina (IFSC)*, São José/SC
Vocational technical course in Telecommunications.

Academic Experience

- 2023–2027 **Doctoral Research: *Pulsed Electric Fields in Continuous Flow***, *UFSC*, Florianópolis/SC
Effective electric dose modeling applied to the inactivation of yeast (*Saccharomyces cerevisiae*) by pulsed electric fields in continuous flow, validated against the author's own experimental data: bench instrumentation, oscilloscope acquisition, parameter estimation and out-of-sample validation.
- 2025 **Teaching Assistantship**, *UFSC*, Florianópolis/SC
Transducers Laboratory course, for the Electrical Engineering and Electronic Engineering programs.
- 2026 **Undergraduate Thesis Committee: *Luiza Alves Bongioiolo***, *UFSC*, Florianópolis/SC
Examiner for the thesis *Contact impedance sensing in electroporation: quantifying the effects of conductive gel anomalies in electrochemotherapy*. Electrical Engineering. Advised by Prof. Daniella de Lourdes Luna Santana de Andrade, Ph.D., and co-advised by Prof. Daniela Ota Hisayasu Suzuki, Ph.D.
- 2026 **Undergraduate Thesis Committee: *Denise Tuda***, *UFSC*, Blumenau/SC
Examiner for the thesis *Safety system design for a hydraulic press brake under NR-12: a retrofit case study based on the ISO 12100 and ISO 13849 standards*. Control and Automation Engineering. Advised by Prof. Guilherme Brasil Pintarelli, Ph.D., and co-advised by Prof. Franciely Velozo Aragão, Ph.D.
- 2025 **Undergraduate Thesis Committee: *João Henrique Anizelli Godoi***, *UFSC*, Florianópolis/SC
Examiner for the thesis *Automated system for detecting and classifying cell viability in *Saccharomyces cerevisiae* using YOLOv8*. Electronic Engineering. Advised by Prof. Raul Guedert, Ph.D., and co-advised by Prof. Daniela Ota Hisayasu Suzuki, Ph.D.
- 2017 **Undergraduate Thesis Committee: *Jéssica Rodrigues da Silva***, *UFSC*, Florianópolis/SC
Examiner for the thesis *Analysis of electrical, geometrical and chemical parameters in the application of nanosecond electric fields to biological cells*. Electronic Engineering.
- 2017 **Undergraduate Thesis Committee: *Gabriel Gonçalves Neves***, *UFSC*, Florianópolis/SC
Examiner for the thesis *In vitro electroporation study: numerical simulation and experiments*. Electronic Engineering.

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- 2014 **Undergraduate Thesis Committee:** *Ana Paula Rosa Negri, UFSC*, São José/SC
Examiner for the thesis *Epileptic seizure detection from electroencephalogram signals using the Wavelet transform*. Telecommunications Systems.
- 2013–2014 **Master's Dissertation:** *Single-Cell Electroporation Study, UFSC*, Florianópolis/SC
Electroporation of isolated biological cells through a capillary electrode: numerical studies of the electrical and mechanical effects on the cell membrane. Advised by Prof. Daniela O. H. Suzuki, Ph.D., and Prof. Jefferson L. B. Marques, Ph.D.
- 2012 **Teaching Assistantship, UFSC**, Florianópolis/SC
Transducers Laboratory course, for the Electrical Engineering and Electronic Engineering programs.

Academic Publications

- 2026 **ANSELMO, J., SUZUKI, D. O. H., PINTARELLI, G. B., ANTÔNIO JÚNIOR, A. de C., SILVA, J. R. da.** Mathematical Modeling and Experimental Assessment of Pulsed Electric Fields in Continuous Flow for *Saccharomyces cerevisiae*. *Submitted, under review*, 2026.
- 2014 **SUZUKI, D. O. H., ANSELMO, J., OLIVEIRA, K. D. de, FREYTAG, J. O., RANGEL, M. M. M., MARQUES, J. L. B., RAMOS, A.** Numerical Model of Dog Mast Cell Tumor Treated by Electrochemotherapy. *Artificial Organs*, 2014.
- 2014 **FRONZA, C. F., ANSELMO, J., MARQUES, J. L. B., PINTARELLI, G. B., CASTRO, A. de.** A Blower-Like Approach to Predict the Effectiveness of Vaccines in a TB Dynamic. *International Journal of Engineering Research and Applications*, Delhi, v. 4, n. 6, p. 233–238, 2014.
- 2014 **FRONZA, C. F., ANSELMO, J., MARQUES, J. L. B.** In silico system for early diagnosis of diabetes mellitus complications by using dynamic pupillometry and heart rate variability. *BIOMAT 2014 — International Symposium on Mathematical and Computational Biology*, Poznan, Poland, 2014.
- 2014 **ANSELMO, J., SUZUKI, D. O. H., MARQUES, J. L. B.** Numerical Studies of the Influential Parameters of Electroporation Through Stretched Microcapillaries. *XXIV Brazilian Congress on Biomedical Engineering — Uberlândia/MG*, 2014.

Patents and Registrations

- 2026 **ANSELMO, Jânio; PINTARELLI, G. B.** Resonant-excitation-based electronic electroporation system (*ElectroRES-100A*). Brazil, 2026. Invention patent application filed through SINOVA — UFSC Innovation Agency, internal UFSC/SINOVA file 2026032378000383. A system in which the treated biological medium is functionally part of the electrical circuit, enabling dynamic impedance monitoring and adaptive control of the electroporation process.
- 2025 **ANSELMO, Jânio.** Electronic process using ultrasound technology for real-time volumetric measurement in liquid waste trucks. Brazil, 2025. Invention patent, application BR 10 2025 001160 3, filed 22 January 2025 with INPI — Brazilian National Institute of Industrial Property. Funding institution: ENSA Tecnologia.

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- 2016 **ANSELMO, Jânio; ANSELMO, T.** Trademark and patent registration for ENSA Tecnologia products and services. Brazil, 2016. Registered trademark 925500895, INPI — Brazilian National Institute of Industrial Property.

Peer Review, Teaching and Outreach

- 2013–2014 Article reviewer for AMIA (*American Medical Informatics Association*), with three reviews completed in 2014.
- 2013 Instructor of the course *Biomedical signal processing* at the *5th Minicourse on Biomedical Engineering in Practice*.
- 2012 Organizer of the *4th Minicourse on Biomedical Engineering in Practice*.

Professional Experience

- 2009–present **Co-founder and R&D Director, ENSA Tecnologia, Palhoça/SC**
Leadership of the research and development area, with a team of four. Responsible for the concept and electronic design of the company's products, from microcontroller *hardware* to embedded algorithms, across healthcare, industrial automation, environmental monitoring and the Internet of Things. Recent work includes the electronic process for ultrasound-based real-time volumetric measurement in liquid waste trucks, filed as a patent in 2025 (see *Patents and Registrations*). The projects listed below were developed in this role.
- 2021 **Home Care Health Device, ENSA Tecnologia, Palhoça/SC, ENSABiofeedback-100A**
Microcontroller-based electronic design for the acquisition of multiple physiological signals: photoplethysmography (PPG), electromyography (EMG) and galvanic skin response (GSR). Device used for *biofeedback* assessment and training.
- 2019 **Environmental Automation Equipment, ENSA Tecnologia, Palhoça/SC, ENSAMed-200S**
Microcontroller-based electronic design for fuel station automation, monitoring environmental sensors in fuel pump *sumps* and fuel tank interstitial spaces (SASC).
- 2017 **Optical Character Recognition (OCR) automation algorithms, ENSA Tecnologia, Palhoça/SC, ENSAPlate**
Development of pattern classification algorithms, involving Artificial Neural Networks (ANN), for applications in fuel stations and gated communities. The system classifies vehicle license plates for fuel pump automation.
- 2017 **Control and Automation Project (Internet of Things), ENSA Tecnologia, Palhoça/SC, ENSAIoT-200A**
Microcontroller-based electronic design for monitoring physical quantities such as volume, pressure, water flow and temperature, with interfaces for *real-time* data transmission — *Software as a Service* (SaaS) — in industrial and residential-complex applications.

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- 2016 **Development Platform (Internet of Things), ENSA Tecnologia, Palhoça/SC, ENSAIoT-100A**
Open source microcontroller board for the development of control and automation projects. The board provides inputs (digital and analog), outputs (digital and analog), *Bluetooth*, *Wi-Fi* and serial (RS-232 and RS-485) transmission interfaces, LCD *display*, keypad (*touch* keys) and electromechanical actuation (automation relays) for residential, industrial and residential-complex applications.
- 2015 **Biometric Access Control, ENSA Tecnologia, Palhoça/SC, ENSAControll-100A**
 Microcontroller-based electronic design with *ethernet* and GSM/GPRS connectivity, for applications in hostile environments (without internet access) and with a centralized database. Used in people management, such as electronic time tracking.
- 2013 **Vital Signs Monitoring Platform, UFSC, Florianópolis/SC, Low Cost Spirometry**
 Development of low-cost *hardware* for monitoring heartbeat and cardiac audio in ubiquitous healthcare settings. Project carried out in the *Smart Medical Devices* course with Prof. Mohamad Sawan, P.Eng., Ph.D.
- 2012 **Entertainment Platform, ENSA Tecnologia, Palhoça/SC, ENSAAirGame-100A**
 Microcontroller-based electronic design of an *air hockey table* for entertainment in commercial venues. The system controls users by token or remotely rechargeable Radio-Frequency Identification (RFID) card.
- 2011 **Volumetric and Environmental Measurement Equipment, ENSA Tecnologia, Palhoça/SC, ENSAMed-100B**
 Electronic design for environmental sensor monitoring and volumetric measurement in fuel pump *sumps* and fuel tank interstitial spaces (SASC).
- 2009 **Monitored Alarm Panel, ENSA Tecnologia, Palhoça/SC, ENSAHome-100A**
 Microcontroller-based electronic design for residential and small-business alarms. The system performs motion detection and human presence monitoring in indoor environments.

Tools

Programming Matlab, Octave, C/C++, Python, Java.
 Architectures Microchip (PIC), Atmel (AVR), Espressif (ESP).
 CAD Eagle, Altium Designer, KiCAD.
 Simulation PSpice, Proteus, COMSOL Multiphysics.
 Documentation $\text{\LaTeX}$, MS-Office, LibreOffice.
 Instrumentation Oscilloscope data acquisition, impedance spectroscopy, conductivity measurement.

Languages

- **Portuguese** — fluent (native);
- **English** — intermediate;
- **Spanish** — basic.

Volunteering

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- 2020–present **Assistant Pastor**, *Centro Evangélico Missões (CEM)*, Palhoça/SC
Leader of the Multimedia Ministry, supervisor of the Growth Groups (GCs), Theology teacher and leader of the Couples Ministry.
- 2019–present **Theology Teacher**, *Centro Evangélico Missões (CEM)*, Palhoça/SC
Courses taught: Discipleship, Bibliology, Evangelism and Growth Groups, Baptism in the Holy Spirit and Spiritual Gifts, Church History, Introduction to the Old Testament, Introduction to the New Testament and Homiletics.
- 2023 **Teacher at the School of Missions**, *Centro Evangélico Missões (CEM)*, Palhoça/SC
Course taught: Geography of Africa.

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